

Sampling Models for Contingency Tables

Presentation Outline

- 1 Background Information
- 2 One Population
- 3 Multiple Populations

Distributions Related to Categorical Variables

- Binomial Distribution
- Multinomial Distribution

Binomial Distribution

Categorical Variable with 2 categories = Success and Failure

p = Proportion (Probability) of Successes in Population

Define: Y = number of Successes in n independent observations

Y is said to have a Binomial Distribution with parameters n and p .

- Mean: $\mu = E(Y) = np$
- Variance: $\sigma^2 = V(Y) = np(1 - p)$

Multinomial Distribution

Categorical Variable with $I > 2$ categories

p_i = Proportion (probability) of observations belonging to category i in population

Define: Y_i = number of observations in the i th category in n independent observations for $i = 1, 2, \dots, I$

$Y_1, Y_2, \dots, Y_I$ is said to have a Multinomial distribution with parameters n and $p_1, p_2, \dots, p_I$.

- Means: $E(Y_i) = np_i$
- Variances: $V(Y_i) = np_i(1 - p_i)$
- Correlations: $\text{Corr}(Y_i, Y_I) = -\sqrt{\frac{p_i p_I}{(1-p_i)(1-p_I)}}$

One Population - Variables and Data Collection

- Variable 1
 - I categories
- Variable 2
 - J categories
- Data Collection and Summary
 - Random sample of size n from population
 - Cross-classify observations according to categories of two variables
 - Form into contingency table

One Population - Contingency Table Example

	Variable 2				
Variable 1	Cat. 1	Cat. 2	Cat. 3	Cat. 4	Total
Cat. 1	Y_{11}	Y_{12}	Y_{13}	Y_{14}	$Y_{1.}$
Cat. 2	Y_{21}	Y_{22}	Y_{23}	Y_{24}	$Y_{2.}$
Cat. 3	Y_{31}	Y_{32}	Y_{33}	Y_{34}	$Y_{3.}$
Total	$Y_{.1}$	$Y_{.2}$	$Y_{.3}$	$Y_{.4}$	n

One Population - Distribution of Y_{ij}

Define a new category variable as the combination of Variables 1 and 2.

This new Categorical Variable has $I * J > 2$ categories

Y_{ij} = number of observations in i^{th} category of Variable 1 and j^{th} category of variable 2 in n independent observations from sample.

Set of $I * J$ observations Y_{ij} has a Multinomial distribution with parameters:

- n observations
- p_{ij} = set of $I * J$ proportions (probabilities) of observations in categories $i = 1, 2, \dots, I$ for Variable 1 and categories $j = 1, 2, \dots, J$ for Variable 2 in the population.
- $E(Y_{ij}) = np_{ij}$ for all i and j .

Multiple Populations - Variables and Data Collection

- Variable 1
 - I categories
- Variable 2
 - $J = 2$ categories
 - $J > 2$ categories
- Data Collection and Summary
 - Stratified samples from I populations or experiment with I treatment groups.
 - Cross-classify observations according to categories of two variables
 - Form into contingency table

Multiple Populations - Contingency Table Example $J = 2$

Variable 1	Variable 2		Total
	Success	Failure	
Cat. 1	Y_{11}	Y_{12}	n_1
Cat. 2	Y_{21}	Y_{22}	n_2
Cat. 3	Y_{31}	Y_{32}	n_3
Total	$Y_{.1}$	$Y_{.2}$	n

Multiple Populations - Distribution of Y_{i1}

We have n_i independent observations from I independent samples.

Y_{i1} = number of successes in n_i independent observations from sample i .

For each i , Y_{i1} has a Binomial distribution with parameters:

- n_i observations
- p_{i1} = proportion (probability) of success in population i
- $E(Y_{i1}) = n_i p_{i1}$ for all i

The I Binomial values Y_{i1} are independent for $i = 1, 2, \dots, I$.

Multiple Populations - Contingency Table Example $J > 2$

Variable 1	Variable 2				Total
	Cat. 1	Cat. 2	Cat. 3	Cat. 4	
Cat. 1	Y_{11}	Y_{12}	Y_{13}	Y_{14}	n_1
Cat. 2	Y_{21}	Y_{22}	Y_{23}	Y_{24}	n_2
Cat. 3	Y_{31}	Y_{32}	Y_{33}	Y_{34}	n_3
Total	$Y_{.1}$	$Y_{.2}$	$Y_{.3}$	$Y_{.4}$	n

Multiple Populations - Distribution of Y_{ij}

We have n_i independent observations from I independent samples.

Y_{ij} = number of observations in category j of Variable 2 in sample i .

For each i , $Y_{i1}, Y_{i2}, \dots, Y_{iJ}$ has a Multinomial distribution with parameters:

- n_i observations
- $p_{i1}, p_{i2}, \dots, p_{iJ}$ = set of proportions (probabilities) for categories $j = 1, 2, \dots, J$ from Variable 2 for population i .
- $E(Y_{ij}) = n_i p_{ij}$ for all i and j

The I sets of Multinomial values $Y_{i1}, Y_{i2}, \dots, Y_{iJ}$ are independent for $i = 1, 2, \dots, I$.